PlasmaMade®
Absolute Science

July 2024

TESTS BY THE DANISH TECHNICAL INSTITUTE ON A PLASMAMADE FILTER

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Background:

On behalf of PlasmaMade B.V. and MT Nordic AS, the Danish Technical Institute in Aarhus has performed a series of tests on a PlasmaMade filter. In these tests, a number of different but realistic tests were performed with the frying of beef. The goal was to establish an independent performance test of our filters and their result regarding their effectiveness against particulate matter.

Conclusions:

Predictable, using an old-fashioned direct exhaust will give a very good result regarding particulate matter. However, with modern regulations and new demands on energy use, direct exhausts are no longer allowed nor desirable.

In the tests, it is clearly visible that a PlasmaMade filter removes more than 80% of the particle sizes PM_{10}, $PM_{2.5}$ and PM_1 in a timespan of approximately 16 minutes. This takes place in the most ideal circumstances: a slightly higher set hood (350 m^3/h) and a ventilation flow of 36 m^3/h. At PlasmaMade we know that recirculation = ventilation. Never recirculate without proper ventilation. The higher efficiency of the filter with a higher hood setting is also explicable: the polluted air passes more often through the filter, but the speed of the air is such that it is possible for the filter to do its work properly, as is shown in the graphics. Regarding the UFP's (Ultra Fine Particles): they are less affected by the filter in this short time, but our research shows that UFP's will eventually (after approximately 45 minutes) be removed by the filter. It is also clearly visible that with an exhaust UFP's are less affected, that will be due to the Brownian motion which especially affects very small particles. https://www.sciencedirect.com/topics/pharmacology-toxicology-and-pharmaceutical-science/brownian-motion

Test Report

REPORT NUMBER:
300-24-026

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Init: ANDN/ASOV
Order No.: 257690

Assignor: PLASMAMADE B.V.
Achthoevenweg 30
7951 SK Staphorst

Item: Plasmamade electostatic filter

Sampling: The sample was forwarded by the assignor.

Period: Testing took place from 23-05-2024 till 28-06-2024.

Method: The method is described on the following page.

Test result: In the table below the particle removal is stated for each test.

	NO RANGE HOOD	EXHAUST 200 M³/H	EXHAUST 350 M³/H	PLASMAMADE FILTER 200 M³/H	PLASMAMADE FILTER 350 M³/H	PLASMAMADE FILTER 200 M³/H AND 36 M³/H EXHAUST	PLASMAMADE FILTER 350 M³/H AND 36 M³/H EXHAUST
PM1	0%	92%	95%	35%	73%	51%	83%
PM2.5	0%	92%	95%	37%	74%	52%	84%
PM10	0%	92%	95%	51%	81%	63%	88%
UFP	0%	56%	69%	6%	26%	17%	27%

Remarks: Charts illustrating the data are presented on the following pages.

Terms: This test was conducted in accordance with international requirements (ISO/IEC 17025:2017) and in accordance with the General Terms and Conditions of Danish Technological Institute. The test results solely apply to the tested item. This analysis report/test report may be quoted in extract only if Danish Technological Institute has granted its written consent.

Place: Hoodlab - Taastrup

Signature: This document is only valid with a digital signature from Danish Technological Institute. The date of issue appears from the digital signature. Approved and signed by:

Asger Søvsø
Head of hoodlab

Andreas Egholm Nielsen
Co-reader

Method:

The test will be conducted in the same facilities that we use for EN61591 and 13141-3, during the test we will have the same disturbing element as described in the in EN13141-3 and no personal will occupy the room. We fry the beef at 180°C, for 16 minutes on one side, this ensures a good but not charred sear.

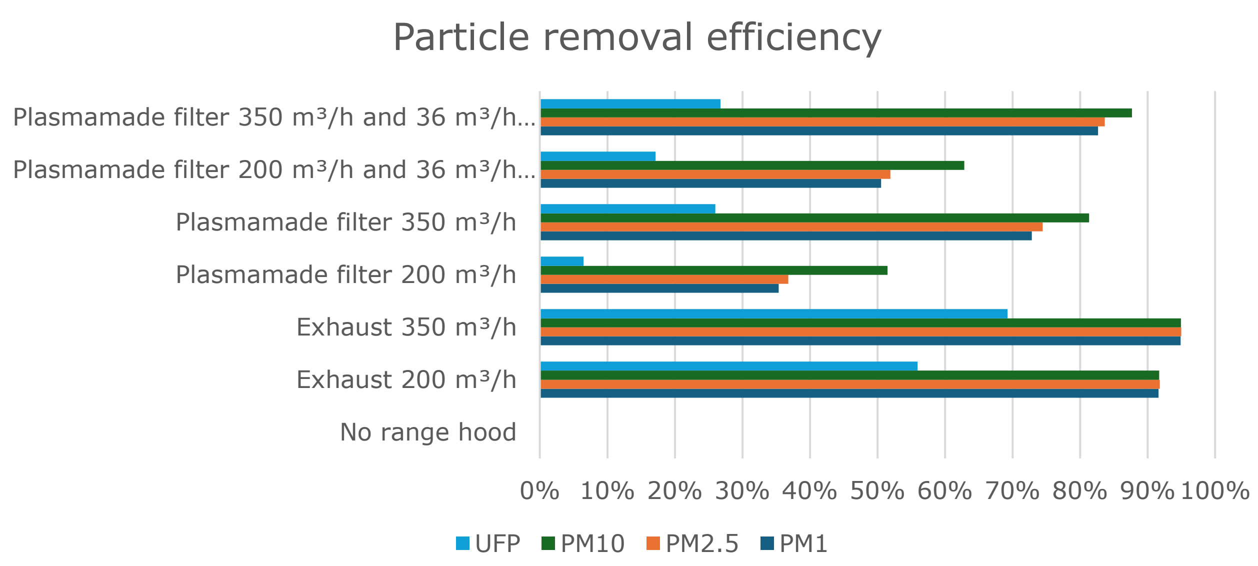
During the test we will measure the particle concentration with a TSI P-trak and dust-trak 1.7m above the floor.

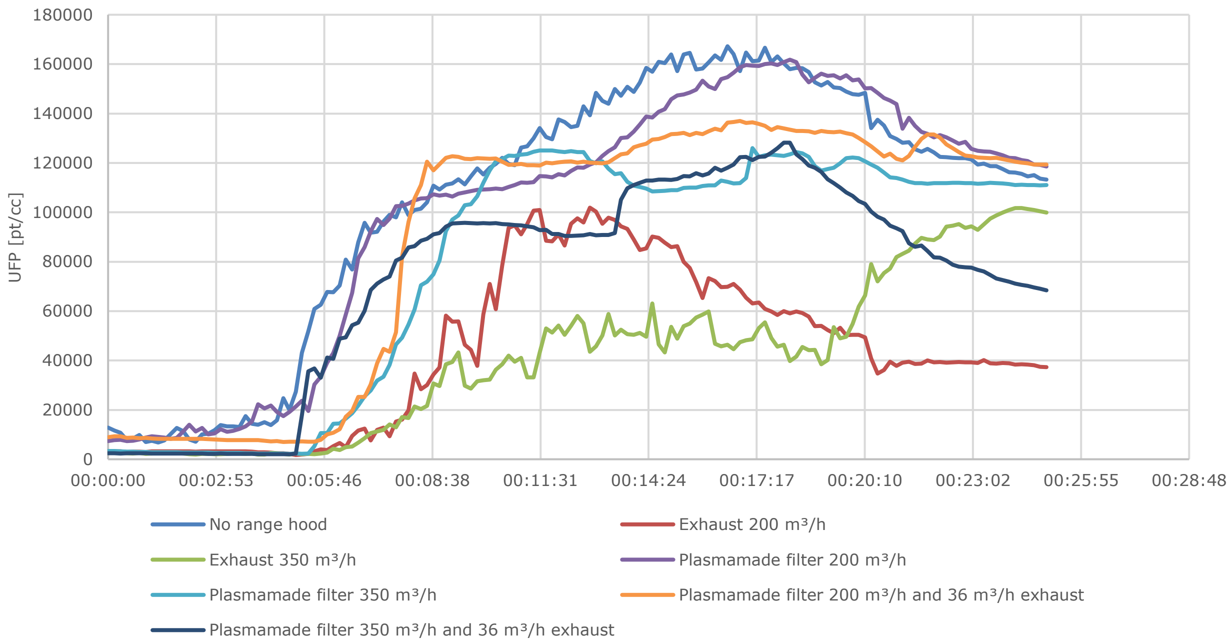
The test is performed at seven different configurations:

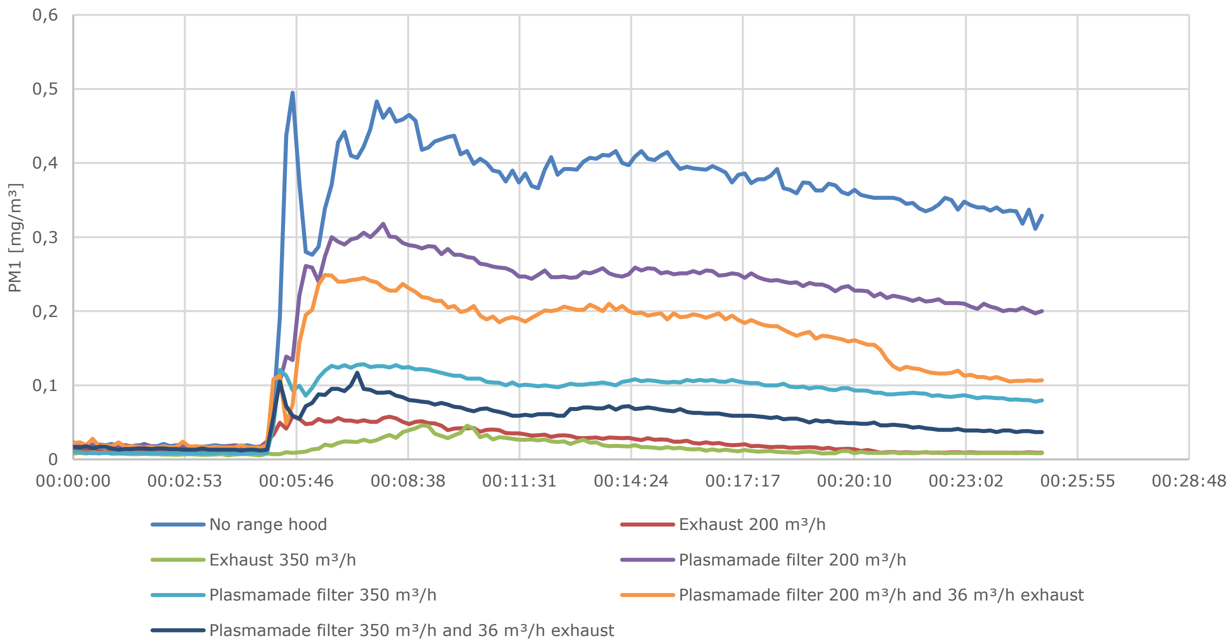
- Without a range hood (baseline)
- Range hood with extraction (350m³/h)
- Range hood with the customer's recirculating filter (350m³/h)
- Range hood with the customer's recirculating filter (350m³/h) and basic ventilation (36m³/h)
- Range hood with extraction (200m³/h)
- Range hood with the customer's recirculating filter (200m³/h)
- Range hood with the customer's recirculating filter (200m³/h) and basic ventilation (36m³/h)

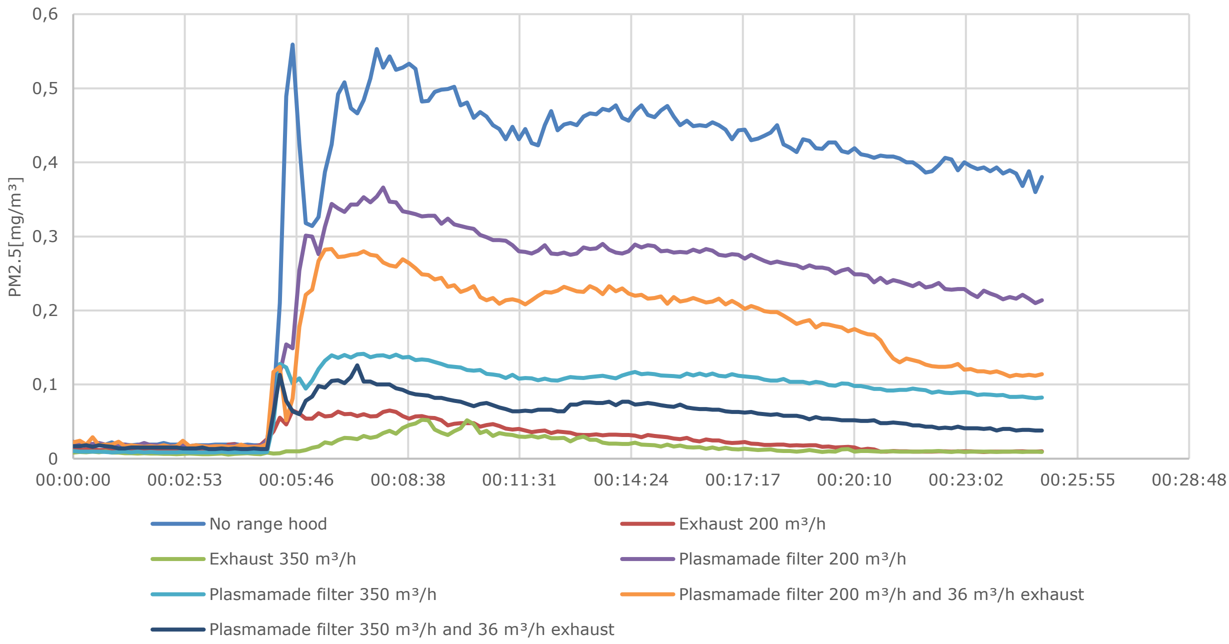
Each test is carried out twice for quality control of the data.

Results:

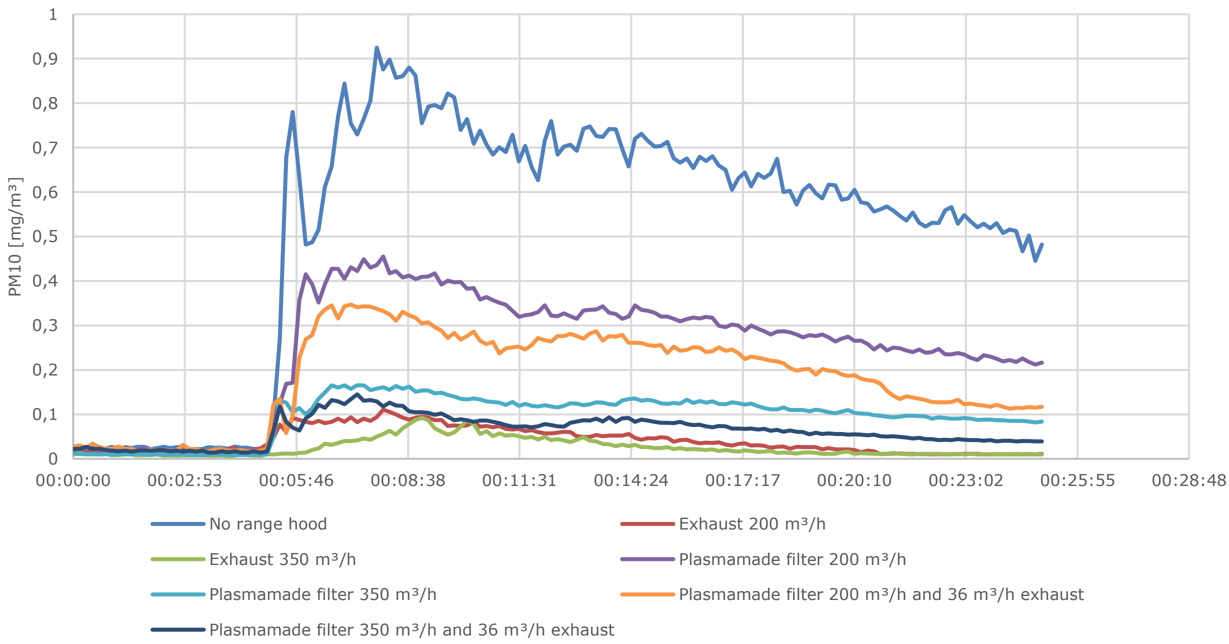



180000
160000
140000
120000
100000
80000
60000
40000
20000
0
UFP [pt/cc]
00:00:00
00:02:53
00:05:46
00:08:38
00:11:31
00:14:24
00:17:17
00:20:10
00:23:02
00:25:55
00:28:48
No range hood
Exhaust 200 m³/h
Exhaust 350 m³/h
Plasmamade filter 200 m³/h
Plasmamade filter 350 m³/h
Plasmamade filter 200 m³/h and 36 m³/h exhaust
Plasmamade filter 350 m³/h and 36 m³/h exhaust


0,6
0,5
0,4
0,3
0,2
0,1
0
PM1 [mg/m³]
00:00:00
00:02:53
00:05:46
00:08:38
00:11:31
00:14:24
00:17:17
00:20:10
00:23:02
00:25:55
00:28:48
No range hood
Exhaust 200 m³/h
Exhaust 350 m³/h
Plasmamade filter 200 m³/h
Plasmamade filter 350 m³/h
Plasmamade filter 200 m³/h and 36 m³/h exhaust
Plasmamade filter 350 m³/h and 36 m³/h exhaust



0,6
0,5
0,4
0,3
0,2
0,1
0
PM2.5[mg/m³]
00:00:00
00:02:53
00:05:46
00:08:38
00:11:31
00:14:24
00:17:17
00:20:10
00:23:02
00:25:55
00:28:48
No range hood
Exhaust 200 m³/h
Exhaust 350 m³/h
Plasmamade filter 200 m³/h
Plasmamade filter 350 m³/h
Plasmamade filter 200 m³/h and 36 m³/h exhaust
Plasmamade filter 350 m³/h and 36 m³/h exhaust



1
0,9
0,8
0,7
0,6
0,5
0,4
0,3
0,2
0,1
0
PM10 [mg/m³]
00:00:00
00:02:53
00:05:46
00:08:38
00:11:31
00:14:24
00:17:17
00:20:10
00:23:02
00:25:55
00:28:48
No range hood
Exhaust 200 m³/h
Exhaust 350 m³/h
Plasmamade filter 200 m³/h
Plasmamade filter 350 m³/h
Plasmamade filter 200 m³/h and 36 m³/h exhaust
Plasmamade filter 350 m³/h and 36 m³/h exhaust